

Elementos de Probabilidades e Teoria de Números

Elementos de Probabilidades - Soluções da Folha 3

	$E[X]$	$Var[X]$	σ_X	$\chi_{0.25}$	$\chi_{0.5}$	$\chi_{0.75}$
1(a) i.	1	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	0	1	1
1(a) ii.	1	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	0	1	1
1(a) iii.	$\frac{161}{36}$	$\frac{2555}{1296}$	$\sqrt{\frac{2555}{1296}}$	3	5	6
1(a) iv.	$\frac{91}{36}$	$\frac{2555}{1296}$	$\sqrt{\frac{2555}{1296}}$	1	2	3
1(a) v.	$\frac{70}{36}$	$\frac{2660}{1296}$	$\sqrt{\frac{2660}{1296}}$	1	2	3
1(a) vi.	7	$\frac{210}{36}$	$\sqrt{\frac{210}{36}}$	5	7	9
1(a) vii.	$\frac{1}{3}$	$\frac{10}{36}$	$\sqrt{\frac{10}{36}}$	0	0	1
4.	$\frac{7}{2}$	$\frac{37}{12}$	$\sqrt{\frac{37}{12}}$	2	4	5

; Primeiro decil: 2; 0.8

2. (a) Sim; (b) Sim
3. (a) 10; 4.4; (b) 10.1; (c) -3; 14.3
4. (a) 25 (valor médio) e 28.75 (variância);
(b) $N = 11$ é o menor
5. (a) 4×0.6^9 ; 4.6×0.6^9 ; 0.4^{10} ;
(b) 45×0.5^{10} , $1 - 11 \times 0.5^{10}$;
(c) igual a (b)
(d) $(\frac{3}{5})^4$; $(\frac{2}{5})^4$; 0 no caso de extracção sem reposição

Exercícios Suplementares à Folha 3

	$E[X]$	$Var[X]$	σ_X	$\chi_{0.25}$	$\chi_{0.5}$	$\chi_{0.75}$
1.	1.	3.15	2.1275	$\sqrt{2.1275}$	2	3
	2.	$\frac{13}{6}$	$\frac{11}{36}$	$\frac{\sqrt{11}}{6}$	$\sqrt{3}$	$\sqrt{5}$
	3.	0	$\frac{1}{6}$	$\frac{\sqrt{6}}{6}$	$-1 + \frac{\sqrt{2}}{2}$	0
						$1 - \frac{\sqrt{2}}{2}$
2.	(a)	$F_X(c) = \begin{cases} 0 & se \ c < 0 \\ c^3(4-3c) & se \ 0 \leq c \leq 1 \\ 1 & se \ c > 1 \end{cases}$; $E[X] = \frac{3}{5}$; $Var[X] = \frac{1}{25}$				
	(b) i.	$L : \begin{cases} \frac{8}{27} & \frac{13}{27} & \frac{18}{27} \end{cases}$; ii. $\frac{391}{27}$				
	(c)	$Y \sim Bin(5, \frac{1}{9})$; $Y : \begin{cases} 0 & 1 & 2 & 3 & 4 & 5 \\ (\frac{8}{9})^5 & 5 \times \frac{8^4}{9^5} & 10 \times \frac{8^3}{9^5} & 10 \times \frac{8^2}{9^5} & 5 \times \frac{8}{9^5} & (\frac{1}{9})^5 \end{cases}$				
3.	(a)	$X : \begin{cases} \frac{1}{3} & \frac{2}{3} & \frac{3}{3} \end{cases}$; $Y : \begin{cases} \frac{2}{3} & \frac{3}{3} \end{cases}$; $E[X] = 2$; $Var[X] = \frac{2}{3}$; $E[Y] = \frac{8}{3}$; $Var[Y] = \frac{2}{9}$				
	(b)	$\frac{1}{4}$ e 0; X e Y não são independentes				
	(c) i.	$S : \begin{cases} \frac{3}{6} & \frac{4}{6} & \frac{5}{6} & \frac{6}{6} \end{cases}$; $T : \begin{cases} \frac{0}{6} & \frac{1}{6} & \frac{2}{6} \end{cases}$; ii. $\frac{14}{3}$; $\frac{2}{3}$; $\frac{11}{9}$; $\frac{5}{9}$				
4.	(a)	—				
	(b)	$N \sim Exp(n\lambda)$				